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$$\begin{aligned} & \int \frac{dx}{3 \sin x + 2 \cos x + 2} \\ t = \tan \frac{x}{2} & \Rightarrow \sin x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}, dx = \frac{2dt}{1+t^2} \\ & = \int \frac{\frac{2}{1+t^2} dt}{3 \left(\frac{2t}{1+t^2} \right) + 2 \left(\frac{1-t^2}{1+t^2} \right) + 2} \\ & = \int \frac{2dt}{6t + 2(1-t^2) + 2(1+t^2)} \\ & = \int \frac{2dt}{6t + 2 - 2t^2 + 2 + 2t^2} \\ & = \int \frac{2dt}{6t + 4} \\ & = \int \frac{dt}{3t + 2} \\ & = \frac{1}{3} \ln |3t + 2| + C \\ & = \frac{1}{3} \ln \left| 3 \tan \frac{x}{2} + 2 \right| + C \end{aligned}$$

References